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WO-300-06 — Replace spool SP03 of 4"-P150-CWS-3003

Area	A300 Exchangers and drum	**Task**	SR Spool replacement
Priority	3 (reliability)	**Issued by**	Facility owner (demonstration)
Primary asset	**E-102** — E-102 shell-and-tube exchanger 36" ID × 192" — E 134'-6" N -25'-0" EL 106'-6"		
Equipment	E-102	**Lines**	4"-P150-CWS-3003
Isometrics	ISO-4"-P150-CWS-3003 (generated); owner's drawings ISO-CWS-3003		
Permits	General work permit, Hot work permit		

Reason

Internal corrosion on the cooling water supply branch.

Scope of work

1. Isolate per the isolation list; drain, vent and purge 4"-P150-CWS-3003 between the isolations; confirm gas-free for hot work.
2. Remove spool 4"-P150-CWS-3003-SP03 (5 items, 39'-7-5/16" of pipe) — cut at the field welds and break the flanged joints at its ends as listed.
3. Fabricate the replacement spool to ISO-4"-P150-CWS-3003 with the same items (parts list), welding to WS-01; 4 shop welds.
4. Install the spool; make 2 field welds; new gaskets and studs at flanged ends, tightened per SW-WO-5.
5. NDE per the NDE table; hydrostatic test at 210 psi for 30 minutes with the relief valves blinded (SW-TEST-1).
6. Remove blinds, restore the line-up, hand back.

Affected items: 4"-P150-CWS-3003/p8, 4"-P150-CWS-3003/el9, 4"-P150-CWS-3003/p10, 4"-P150-CWS-3003/el11, 4"-P150-CWS-3003/p12

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-1, SW-NDE-2, SW-NDE-4, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-4, SW-WO-5, SW-WO-7, SW-WPS-1, SW-WPS-2, SW-WPS-3

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-CWS-3003	4"-P150-CWS-3003	4"-P150-CWS-3003	1	agrees with the model

ISO-CWS-3003 (PDF) — the generated isometrics follow the ITP.

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
E-102	equipment	E-102 shell-and-tube exchanger 36" ID × 192"		E 134'-6" N -25'-0" EL 106'-6"
4"-P150-CWS-3003/p8	item	PIPE 4" P150 BEVELLED ENDS	4"-P150-CWS-3003	E 136'-6" N 0.000" EL 125'-10-3/16"
4"-P150-CWS-3003/el9	item	ELBOW 90° R = 5/3 OD 4" P150	4"-P150-CWS-3003	E 136'-6" N -3-5/16" EL 126'-2-11/16"
4"-P150-CWS-3003/p10	item	PIPE 4" P150 BEVELLED ENDS	4"-P150-CWS-3003	E 136'-6" N -12'-6" EL 126'-6"
4"-P150-CWS-3003/el11	item	ELBOW 90° R = 5/3 OD 4" P150	4"-P150-CWS-3003	E 136'-6" N -24'-8-11/16" EL 126'-2-11/16"
4"-P150-CWS-3003/p12	item	PIPE 4" P150 BEVELLED ENDS	4"-P150-CWS-3003	E 136'-6" N -25'-0" EL 118'-2-3/16"
4"-P150-CWS-3003/A3	item	VALVE, GATE, WEDGE, OS&Y 4" P150	4"-P150-CWS-3003	E 136'-6" N 0.000" EL 124'-7-7/8"
4"-P150-CWS-3003/A6	item	FLANGE WELD-NECK 4" P150 RAISED FACE	4"-P150-CWS-3003	E 136'-6" N 0.000" EL 125'-6-5/16"
4"-P150-CWS-3003/B2	item	VALVE, GATE, WEDGE, OS&Y 4" P150	4"-P150-CWS-3003	E 136'-6" N -25'-0" EL 109'-3-13/16"
4"-P150-CWS-3003/B5	item	FLANGE WELD-NECK 4" P150 RAISED FACE	4"-P150-CWS-3003	E 136'-6" N -25'-0" EL 110'-2-5/16"
4"-P150-CWS-3003/W2	weld	Butt weld 4" P150 (field)	4"-P150-CWS-3003	E 136'-6" N 0.000" EL 125'-9-1/16"
4"-P150-CWS-3003/W3	weld	Butt weld 4" P150 (shop)	4"-P150-CWS-3003	E 136'-6" N 0.000" EL 125'-11-5/16"
4"-P150-CWS-3003/W4	weld	Butt weld 4" P150 (shop)	4"-P150-CWS-3003	E 136'-6" N -6-11/16" EL 126'-6"
4"-P150-CWS-3003/W5	weld	Butt weld 4" P150 (shop)	4"-P150-CWS-3003	E 136'-6" N -24'-5-5/16" EL 126'-6"
4"-P150-CWS-3003/W6	weld	Butt weld 4" P150 (shop)	4"-P150-CWS-3003	E 136'-6" N -25'-0" EL 125'-11-5/16"
4"-P150-CWS-3003/W7	weld	Butt weld 4" P150 (field)	4"-P150-CWS-3003	E 136'-6" N -25'-0" EL 110'-5"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P150-CWS-3003/A3	4"-P150-CWS-3003	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWS-3003/A3)	E 136'-6" N 0.000" EL 124'-7-7/8"		
2	blind	4"-P150-CWS-3003/A6	4"-P150-CWS-3003	Fit a line blind between 4"-P150-CWS-3003/A3 and FLANGE WELD-NECK 4" P150 RAISED FACE (4"-P150-CWS-3003/A6), work side	E 136'-6" N 0.000" EL 125'-3-5/8"		
3	valve	4"-P150-CWS-3003/B2	4"-P150-CWS-3003	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWS-3003/B2)	E 136'-6" N -25'-0" EL 109'-3-13/16"		
4	blind	4"-P150-CWS-3003/B5	4"-P150-CWS-3003	Fit a line blind between 4"-P150-CWS-3003/B2 and FLANGE WELD-NECK 4" P150 RAISED FACE (4"-P150-CWS-3003/B5), work side	E 136'-6" N -25'-0" EL 109'-11-9/16"		

Parts

Description	Qty	Replaces
PIPE 4" P150 BEVELLED ENDS	3	
ELBOW 90° R = 5/3 OD 4" P150	2	

New welds (SW-WO-4)

Weld	Replaces	On drawing	Line	Size	Wall	Shop/field	Spec	NDE	Location	Welder	Date
WO-300-06-W01	4"-P150-CWS-3003/W2	ISO-CWS-3003 sh 1 W2	4"-P150-CWS-3003	4" P150	0.1875 in	field	WS-01	VT + RT	E 136'-6" N 0.000" EL 125'-9-1/16"		
WO-300-06-W02	4"-P150-CWS-3003/W3	ISO-CWS-3003 sh 1 W3	4"-P150-CWS-3003	4" P150	0.1875 in	shop	WS-01	VT	E 136'-6" N 0.000" EL 125'-11-5/16"		
WO-300-06-W03	4"-P150-CWS-3003/W4	ISO-CWS-3003 sh 1 W4	4"-P150-CWS-3003	4" P150	0.1875 in	shop	WS-01	VT	E 136'-6" N -6-11/16" EL 126'-6"		
WO-300-06-W04	4"-P150-CWS-3003/W5	ISO-CWS-3003 sh 1 W5	4"-P150-CWS-3003	4" P150	0.1875 in	shop	WS-01	VT	E 136'-6" N -24'-5-5/16" EL 126'-6"		
WO-300-06-W05	4"-P150-CWS-3003/W6	ISO-CWS-3003 sh 1 W6	4"-P150-CWS-3003	4" P150	0.1875 in	shop	WS-01	VT	E 136'-6" N -25'-0" EL 125'-11-5/16"		
WO-300-06-W06	4"-P150-CWS-3003/W7	ISO-CWS-3003 sh 1 W7	4"-P150-CWS-3003	4" P150	0.1875 in	field	WS-01	VT	E 136'-6" N -25'-0" EL 110'-5"		

Welding specification WS-01 — Butt welds, pipe grade, wall to 1/2 in

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Scope	Pipe, fittings and weld-neck flanges of Grade 35 pipe / Grade 36 forging, wall up to 1/2 in
Joint	Single-V, 60° included, root opening 1/8 in, root face 1/16 in (SW-WD-4)
Process	Root: gas-shielded tungsten arc. Fill and cap: shielded metal arc.
Filler	Root SW-F1 wire; fill and cap SW-F2 covered electrode (matching strength, low hydrogen)
Preheat	≥ 30 °F above dew point and ≥ 60 °F, 3 in each side
Interpass	≤ 450 °F
Post-weld heat treatment	Not specified (owner to state)
Rules	SW-WPS-1, SW-WPS-2, SW-WPS-3, SW-WD-4

NDE

Method	Extent	Welds	Basis
VT	100 % of new welds, before and after welding	WO-300-06-W01, WO-300-06-W02, WO-300-06-W03, WO-300-06-W04, WO-300-06-W05, WO-300-06-W06	SW-NDE-1
RT (UT where RT is impractical)	10 % of new welds, at least one per welder (class W150); welds shown are preliminary, final choice by the owner's inspector	WO-300-06-W01	SW-NDE-2

Test — Hydrostatic test

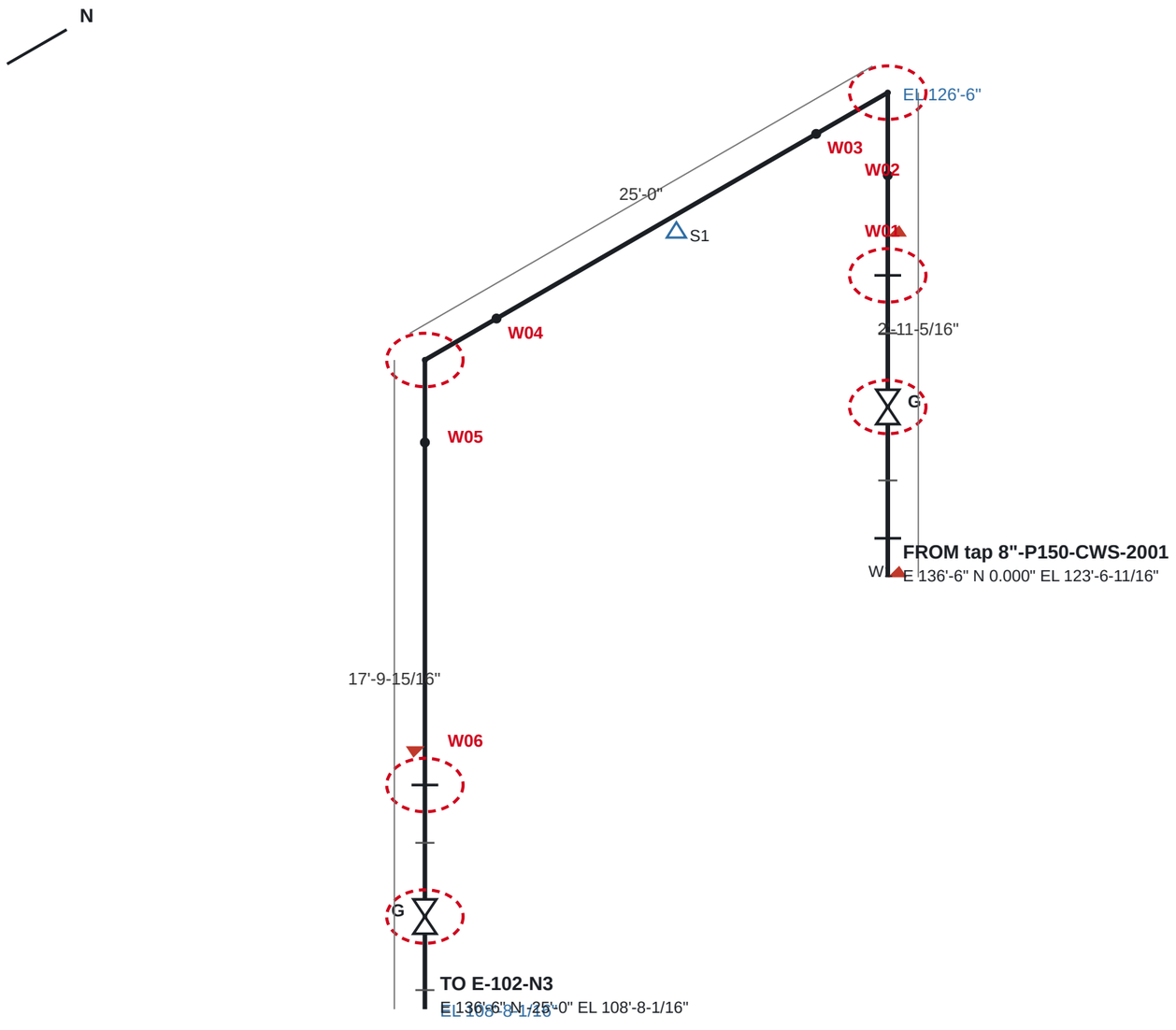
Description	Test boundary between the isolations; relief valves blinded or removed
Pressure	210 psi
Hold	30 minutes, no leak, no pressure drop
Basis	SW-TEST-1

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Line drained, vented and purged; gas-free for hot work	Owner hot-work procedure	Gas test within owner limits	Gas test record, hot work permit	H	H	—				
3	Replacement materials verified against the parts list (size, class, grade, heat marking)	Parts list, SW-LC-3	All items match the line class	Material receiving record	R	R	R				
4	Welding specification WS-01 and welder identification available at the work	WS-01, SW-WPS-3	Specification on site; every welder identified	Welder list	R	R	R				
5	Fit-up of every new weld: root opening and alignment within the root face	SW-WPS-3, SW-WD-4	Root opening 1/8 in; bores aligned within 1/16 in	Fit-up record per weld	H	W	S				
6	Preheat and interpass temperatures maintained during welding	WS-01, SW-WPS-2	Preheat ≥ 30 °F above dew point and ≥ 60 °F, 3 in each side; interpass ≤ 450 °F	Weld record per weld (welder, temperatures)	S	S	S				
7	Visual examination of every new weld	SW-NDE-1, SW-NDE-4	No crack, lack of fusion or incomplete penetration	VT report	W	W	W				
8	Volumetric examination per the NDE table; repairs re-examined	SW-NDE-2, SW-NDE-4	Acceptance per SW-NDE-4 and the owner's criteria	RT/UT reports	R	H	R				
9	Flanged ends: faces inspected, new gaskets, studs to preload	SW-NDE-5, SW-WO-5	No radial scratch; preload as stated	Bolt-up record	W	W	—				
10	Hydrostatic test at 210 psi, 30 minutes	SW-TEST-1	No leak, no pressure drop	Test record with gauge calibration	H	H	H				
11	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
12	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 4"-P150-CWS-3003 — Replace spool SP03 of 4"-P150-CWS-3003



BILL OF MATERIALS

1	FLANGE WELD-NECK 4" P150 RAISED FACE	3
2	GASKET 4" P150 COMPRESSED SHEET	4
3	VALVE, GATE, WEDGE, OS&Y 4" P150	2
4	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150	4
5	PIPE 4" P150 BEVELLED ENDS	39'-7-5/16"
6	ELBOW 90° R = 5/3 OD 4" P150	2

WELDS 7 (3 FIELD) · SUPPORTS 1 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7F

NEW WELDS (WO-300-06)

WO-300-06-W01 replaces W2
WO-300-06-W02 replaces W3
WO-300-06-W03 replaces W4
WO-300-06-W04 replaces W5
WO-300-06-W05 replaces W6
WO-300-06-W06 replaces W7

4"-P150-CWS-3003

CLASS W150 — Water (cooling and process)
SERVICE CWS · 4" · P150 · DESIGN 150 psi
tap 8"-P150-CWS-2001 → E-102-N3
PIPE 39'-7-5/16" · WEIGHT 693 lb

DWG ISO-4"-P150-CWS-3003 REV 0 WO-300-06

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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